

Replacing Softmax with Sparsemax in Transformer Attention

Abstract

The Transformer architecture has become a foundational model in modern deep learning, primarily due to its attention mechanism. A key component of this mechanism is the softmax function, which converts attention scores into probability distributions. However, softmax produces dense attention, assigning non-zero weights to all tokens, which can reduce interpretability. In this project, we replace the softmax function with Sparsemax, a sparse alternative, to investigate its effect on attention behavior and model performance. Experimental results show that Sparsemax induces high sparsity (~88%) in attention weights, improving interpretability while introducing a slight reduction in accuracy (~2–3%), highlighting a trade-off between selective attention and information retention.

1. Introduction

Transformers rely heavily on the self-attention mechanism to model relationships between tokens in a sequence. The standard attention formulation uses the softmax function to normalize attention scores. While effective, softmax produces dense distributions where every token contributes to the final representation.

This project explores whether replacing softmax with Sparsemax can improve interpretability by enforcing sparsity in attention distributions. The goal is to analyze how this modification affects model behavior and performance.

2. Background

2.1 Attention Mechanism

The attention function is defined as:

$$[\text{Attention}(Q, K, V) = \text{softmax}((Q K^T) / \sqrt{d_k}) V]$$

where:

- (Q) , (K) , (V) are query, key, and value matrices
- (d_k) is the dimension scaling factor

Softmax ensures that attention weights are positive and sum to 1.

2.2 Limitations of Softmax

- Produces dense attention (no zero values)
 - Difficult to interpret due to smooth distributions
 - Assigns importance to irrelevant tokens
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2.3 Sparsemax

Sparsemax is an alternative normalization function that produces sparse probability distributions. Unlike softmax, it can assign exact zero values to less important elements, enabling selective attention.

3. Proposed Method

In this project, the softmax function in the attention layer is replaced with Sparsemax:

Original Attention

```
[  
{softmax}(scores)  
]
```

Modified Attention

```
[  
{sparsemax}(scores)  
]
```

This modification changes the behavior of the attention mechanism from dense aggregation to selective filtering.

4. Implementation

- Platform: Google Colab
- Framework: PyTorch
- Model: Simple Transformer with:
 - Embedding layer
 - Single attention layer
 - Fully connected classifier
- Dataset: Synthetic sequence dataset for binary classification

Two models were trained:

1. Transformer with Softmax
 2. Transformer with Sparsemax
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5. Results

5.1 Accuracy Comparison

- Softmax Accuracy: **56.2%**
- Sparsemax Accuracy: **54.0%**

5.2 Sparsity Analysis

- Softmax Sparsity: **0%**
 - Sparsemax Sparsity: **88%**
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6. Discussion

The results highlight a clear behavioral difference between softmax and Sparsemax:

- Softmax produces dense attention distributions, allowing all tokens to contribute.
- Sparsemax enforces sparsity, eliminating approximately 88% of attention weights.

This leads to:

Advantages of Sparsemax

- Improved interpretability
- Clear focus on relevant tokens
- Reduced noise in attention

Limitations

- Slight drop in accuracy (~2–3%)
 - Potential loss of weak but useful signals
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7. Key Insight

Softmax acts as a **soft aggregation mechanism**, combining information from all tokens, whereas Sparsemax behaves as a **hard selection mechanism**, focusing only on the most relevant inputs. This fundamentally changes the inductive bias of the model.

8. Conclusion

This project demonstrates that replacing softmax with Sparsemax significantly alters the behavior of the attention mechanism. Sparsemax introduces high sparsity (~88%), leading to more interpretable attention patterns. However, this comes at the cost of a slight reduction in performance.

The study highlights an important trade-off in deep learning models:

Interpretability vs Performance

9. Future Work

- Explore Entmax for controlled sparsity
- Apply method on real-world datasets (e.g., NLP tasks)

- Extend to multi-head attention
 - Analyze impact on larger Transformer architectures
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10. References

1. Vaswani et al., *Attention Is All You Need*, 2017
 2. Martins & Astudillo, *From Softmax to Sparsemax*, 2016
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